Host Forensics and examination approaches I

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Learning objectives

- Explain the role of file system metadata in digital investigations
- Identify common file signatures (headers and footers) and explain how they are used to determine file type independently of file extensions.
- Describe the process of file carving, including how files may be reconstructed from unallocated space in the absence of file system metadata.
- Assess the evidential value and limitations of metadata and carved files

Identify all files of the system

1. Existing files
2. Deleted, remaining
3. Hidden
4. Encrypted / password protected
   - practical approaches on overcoming encryption
5. Temporary files/ folders
   - File shredders
   - Anti-forensic techniques

Deleted files

- They should be recovered in order to examine their content
- E.g., a hex editor can be used to identify them.

Slack / Unallocated space

- The use of special tools can help the investigator examine the slack and unallocated space of the medium
- Information such as network logon names and passwords can be stored there

Hidden partitions

- Created mainly to store files that the user does not desire to be visible in the operating system.
- illicit data can be missed if the hidden files are not viewed
- A hidden partition may be or may not be installed by the manufacturer.
  - Hidden partitions should be recovered and examined.
  - E.g. Norton Partition Magic can uncover a hidden partition.

Data categories and layers

- Obvious
  - Operating system files
  - Files created by using a specific program
- Obscure
  - May be hidden by the user or by default
  - Files with no obvious meaning or use, e.g. .tmp files
- Hidden
  - Hidden by the user
  - Steganography
- Deleted
  - Deliberately deleted by the user or program
- File residues
  - Ambient data
  - File Slack
  - Free space
  - Shadow data

Active Data

- Data tracked by the file system, still visible in directories and available to applications.
- may be readily comprehensible using simple translation techniques or viewed within an application.
- may be password protected or encrypted
- includes system data residing within the operating system and other obscure data caches.

Latent Data

- Not tracked by the file system, nor readily available
- generally inaccessible without specialized tools and techniques.

Archival Data

- data that's been transferred or backed up to peripheral media
- Archival data can be voluminous, particularly in a large organisation employing frequent, regular back up procedures.
- never reflects all the data that can be identified and extracted - don't carry forward latent data.

NTFS and Windows OS

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All the data in the file system that describes the layout and attributes of the regular files and directories.

Metadata

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Metadata

"Data about data"

- Based on the Greek word "μετά" which means "after"

- Metadata can be the following:
  - The date and time the file was created, accessed, modified, printed
  - The name of the author
  - Organisation name
  - File Size
  - Document version
- This information can be linked and corroborated with other evidence in order to reach to a logical conclusion.

All the data in the file system that describes the layout and attributes of the regular files and directories

Check: Iraq Dossier 2003

Windows View vs NTFS Reality

NTFS stores files as **records and attributes**

Windows presents a **user-friendly view** of files

User actions result in **metadata changes**, not immediate data removal

Types of metadata

System metadata

- NTFS is an example of a system metadata
- information automatically created by the system, when a file is created or copied/moved or updated
- track file location and store demographics about each file, e.g. name, size, creation, modification and usage
- Varies depending on the type of the OS

Application Metadata

- Further information that describes the file and is embedded in the specific file
- Moves with the file when the file is moved or copied to a new location
- Holds information about the file such as tracked changes, document author, document version etc.
- $STANDARD_INFORMATION and $FILE_NAME

•Master File Table (MFT)
•File attributes
•Allocation information
•NTFS logs and journals

•File name
•File size
•Timestamps
•File location

File system metadata benefits

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The information is automatically collected and stored by the system

- all the information that is available to the system is available to be recorded.
- collected automatically with no extra cost for setting up logging mechanisms.
- not necessary to correlate various system logs to obtain the desired information.

Tampering with the information is not as simple as tampering with a file.

- If raw disk access is not allowed by the operating system, the recorded information is protected from all users.
- If raw disk access is allowed a malicious user still has to navigate the file system to get to the information.

1. Context
2. Timelines
3. Persistence beyond deletion
4. Corroboration
5. Activity reconstruction

File Deletion

Deletion - the Windows perspective

- File disappears from folder
- Disk space appears freed
- File may go to Recycle Bin

Deletion – the NTFS impact

- MFT entry marked as unused
- Clusters marked as available
- File content not immediately erased
- Metadata may persist

- MFT records
- File names
- Timestamps
- Partial or full file content

User Action (Windows)

Create / Modify / Delete file

▼

Windows Explorer / OS

Updates file information

▼

NTFS File System

- MFT records
- File attributes
- Allocation status

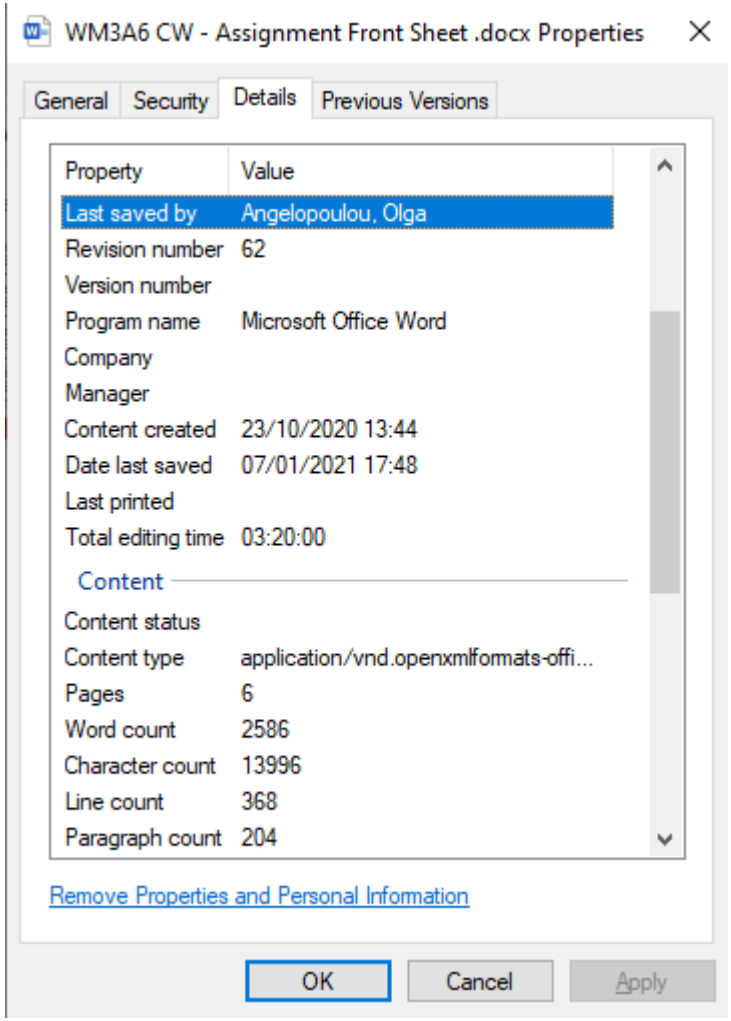
▼

Physical Disk

- Clusters / sectors
- File content
- Unallocated space

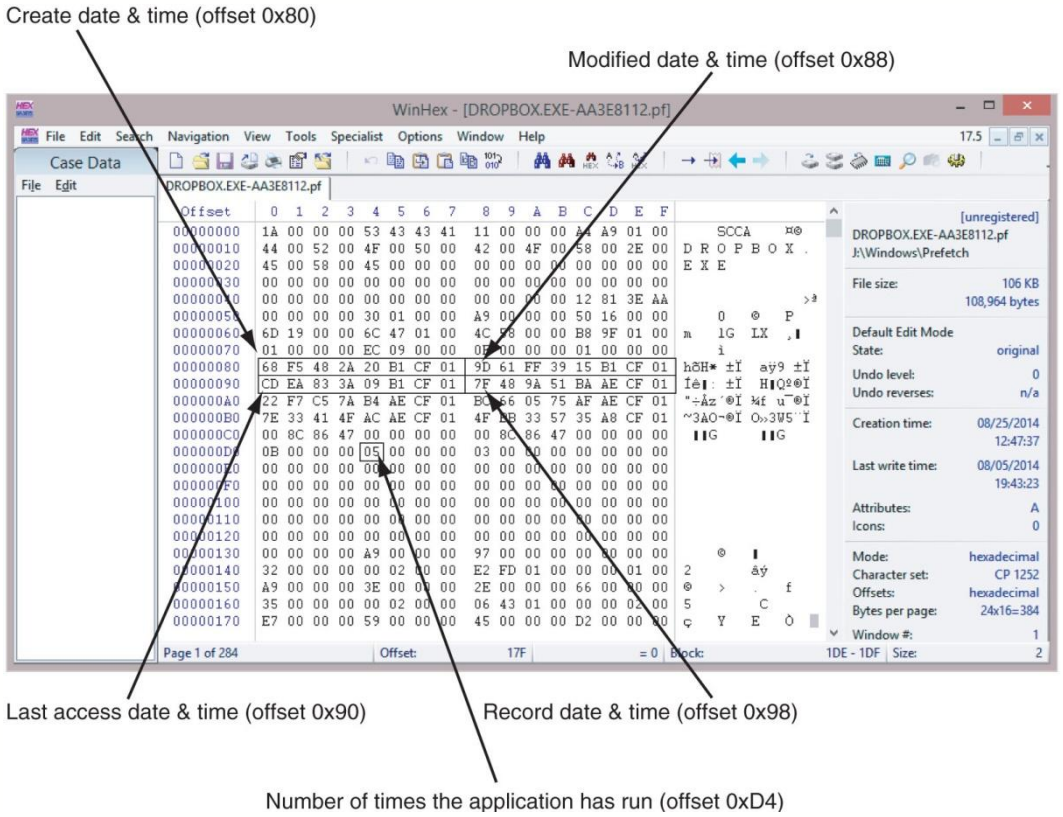
Document (Microsoft) Metadata

- name and initials (or those of the person that created the file)
- organization name
- name of the computer
- the name of the local hard drive or network server where the document was saved
- the name and type of the printer that the document was printed on
- other file properties and summary information -
  - names of previous 10 document authors
  - document revisions, including deleted text that is no longer visible on the screen
  - document versions
  - information about any template used to create the file
  - hidden text and comments.



Prefetch Files

- Prefetch files
- The OS reads the associated prefetch file and loads its information into the computer's memory
- The OS can handle other tasks instead of waiting for an application to load needed libraries
- Useful metadata:
  - application's MAC times in UTC format
  - counter of how many times the app has run



Metadata Anti-Forensics

- Timestamps and other metadata can be easily 'forged' and/ or removed, e.g. TimeStomp

- Common Sense should be used when relying on metadata as a source of evidence
  - Metadata removal utilities, e.g. iScrub, Metadata Assistant, Workshare protect, Doc Scrubber, 3BClean etc

Metadata in digital photographs

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- Raw file format
- biggest disadvantage is that it's proprietary
- not all image viewers can display these formats
- the process of converting raw picture data to another format is referred to as demosaicing
- Exchangeable Image File format (Exif)
- Commonly used to store digital pictures
- Developed by JEITA as a standard for storing metadata in JPEG and TIF files
- Investigators can learn more about the type of digital device and the environment in which photos were taken
- Exif JPEG file's metadata requires special programs, e.g., Exif Reader
- Exif file stores metadata at the beginning of the file

- The metadata section of a JPEG file header can contain information about:
- make and model of the digital camera
- time and date the picture was taken
- distance the camera was focused at
- location information (GPS) where the picture was taken
- small preview image (thumbnail) of the picture
- firmware version, serial numbers, name and version of image manipulation program, etc. …
- Several software tools support the removal of metadata
- jhead (free open source, command line tool)
- Irfanview

Data carving

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Bytes generally stored at the beginning of a file, used by an application for identity of a file type or to identify the content of a file.

File Signatures

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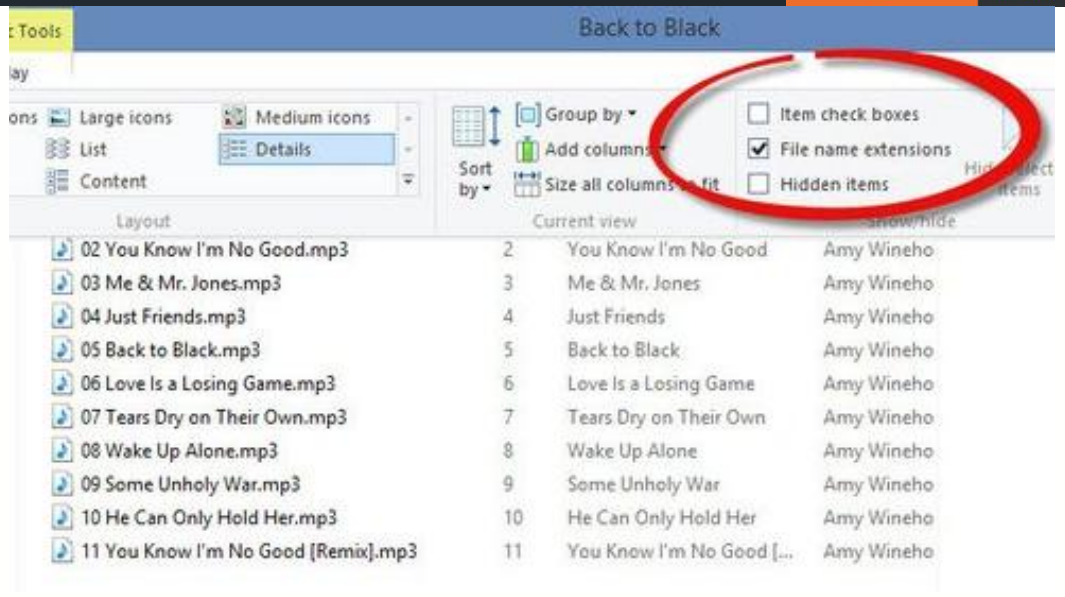
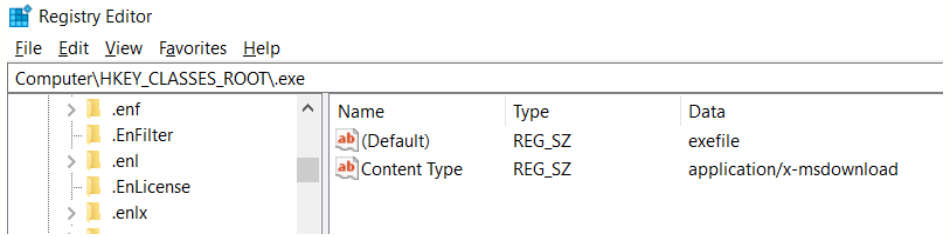
File extension association – Windows OS

- Windows OS has a list of associations between file extensions and applications
- An extension is bound to an application
- All file extensions on a Windows system can be found in the registry hive `HK_CLASSES_ROOT`





File signatures

aka **magic number**

a sequence of characters located within the first 20 bytes of a file, used by an application for identity of a file type or to identify the content of a file.

- *The selection of bytes for signature is arbitrary or application specific*
- Used by application programs to confirm file data before loading and processing the rest of the file.
- File headers/ signatures are associated with specific file extensions.
  - Some users tend to change the file extensions to hide the true identity of the files.
  - E.g. picture.jpg -> picture.doc

To see the file signature, choose an executable file on Windows, such as cal.exe, the executable image for the Calculator on Windows systems, and open it in Notepad. You can use the following command:

```
C:\WINDOWS\system32>notepad calc.exe
```

Notice the characters "MZ" highlighted in the upper left-hand portion of the image.
Executable files ending in .sys, .ocx, .exe, .dll, and .drv will have this signature.

```
FF D8 FF E0 xx xx 4A 46          ÿØÿà..JF
49 46 00                         IF.
                JFIF, JPE, JPEG, JPG  JPEG/JFIF graphics file
                                      Trailer: FF D9 (ÿÙ)

FF D8 FF E1 xx xx 45 78          ÿØÿá..Ex
69 66 00                         if.
                            JPG   Digital camera JPG using Exchangeable Image File Format (EXIF)
                                  Trailer: FF D9 (ÿÙ)
                                  See "Using Extended File Information (EXIF) File Headers in Digital
                                  Evidence Analysis" (P. Alvarez, IJDE, 2(3), Winter 2004) and
                                  ExifTool Tag Names

FF D8 FF E8 xx xx 53 50          ÿØÿè..SP
49 46 46 00                      IFF.
                            JPG   Still Picture Interchange File Format (SPIFF)
                                  Trailer: FF D9 (ÿÙ)
```

NOTES on JPEG file headers: It appears that one can safely say that all JPEG files start with the three hex digits 0xFF-D8-FF. The fourth digit is also indicative of JPEG content. Various options include:

- 0xFF-D8-FF-DB — Samsung D807 JPEG file.
- 0xFF-D8-FF-E0 — Shown above. Standard JPEG/JFIF file.
- 0xFF-D8-FF-E1 — Shown above. Standard JPEG/Exif file.
- 0xFF-D8-FF-E2 — Canon EOS-1D JPEG file.
- 0xFF-D8-FF-E3 — Samsung D500 JPEG file.
- 0xFF-D8-FF-E8 — Shown above. Still Picture Interchange File Format (SPIFF).

File Signatures

Hex Editor Examples

EnCase Setup (x64) 8.08.exe

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00  MZ..........ÿÿ..
00000010  B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00  ¸.......@.......
00000020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000030  00 00 00 00 00 00 00 00 00 00 00 00 40 01 00 00  ............@...
00000040  0E 1F BA 0E 00 B4 09 CD 21 B8 01 4C CD 21 54 68  ..º..´.Í!¸.LÍ!Th
```

MZ = executable file

developed by **M**ark **Z**bikowski

FileSystemAnalysis.pdf

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  25 50 44 46 2D 31 2E 36 0D 25 E2 E3 CF D3 0D 0A  %PDF-1.6.%âãÏÓ..
00000010  31 20 30 20 6F 62 6A 3C 3C 2F 4D 65 74 61 64 61  1 0 obj<</Metada
00000020  74 61 20 32 31 38 38 20 30 20 52 2F 50 61 67 65  ta 2188 0 R/Page
00000030  73 20 33 20 30 20 52 2F 54 79 70 65 2F 43 61 74  s 3 0 R/Type/Cat
```

PDF File

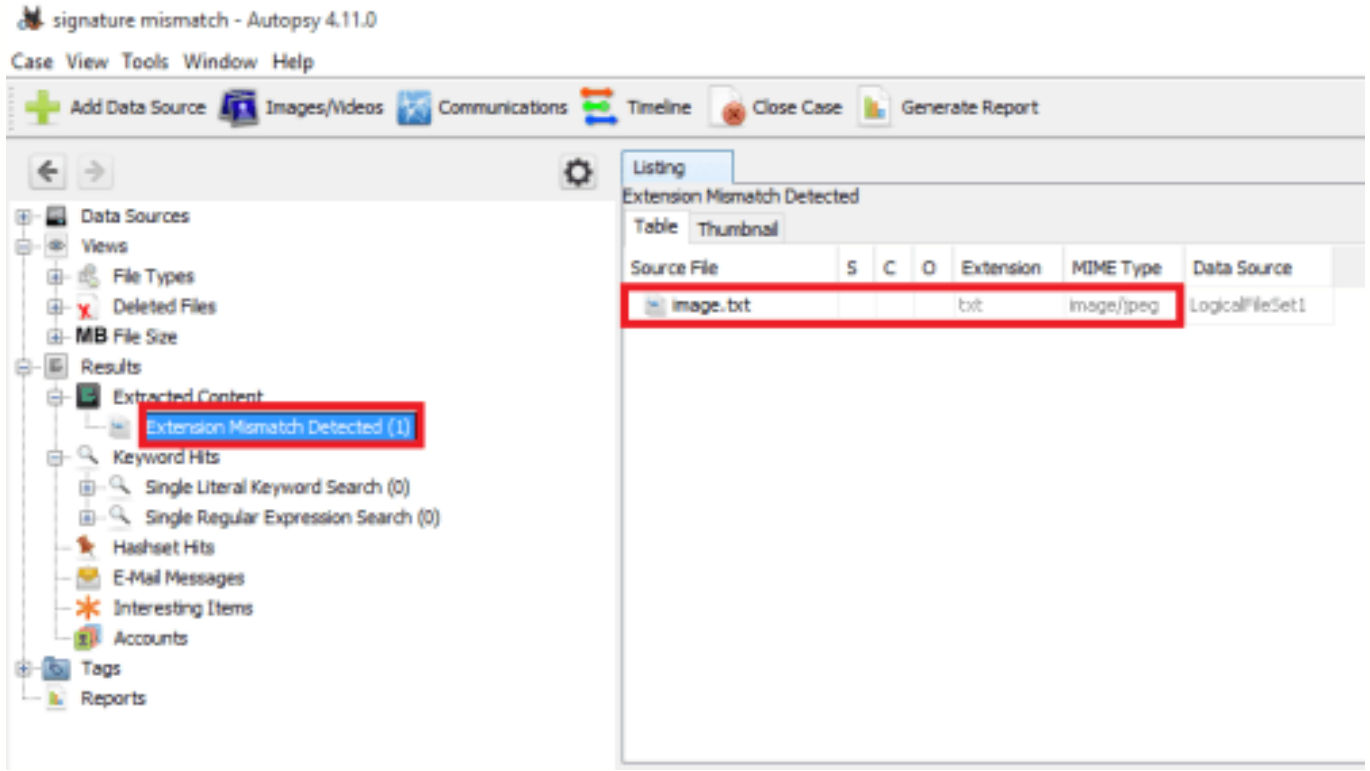
Advanced Digital Forensic Investigation_spec.docx

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F  Decoded text
00000000  50 4B 03 04 14 00 06 00 08 00 00 00 21 00 F0 21  PK..........!.ð!
00000010  EC 7D 8E 01 00 00 13 06 00 00 13 00 08 02 5B 43  ì}Ž...........[C
00000020  6F 6E 74 65 6E 74 5F 54 79 70 65 73 5D 2E 78 6D  ontent_Types].xm
00000030  6C 20 A2 04 02 28 A0 00 02 00 00 00 00 00 00 00  l ¢..( .........
00000040  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

Docx File

File signature analysis

1. detects disparities between the file's signature and extension
2. a user can alter the signature of a file to hide data
3. Forensic tools highlight extension mismatches



File content identification

- A file header contains metadata about the file to help a software application validate the file content is correct, load the data and view or process the data.
- An important element is a **magic number** typically at the start of the file and present in files of that type.
  - Magic number allows a program to initially determine if the file is valid by first checking presence of number.
  - However magic numbers are not rigorously defined by any organisation.

"… the process of extracting a collection of data from a larger data set."

File Carving

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File carving

- Carving searches for artefacts based on content rather than on metadata
- Technique commonly used within digital forensics to find file data
- Close relation to file signatures
  - Involves search blocks of data for file signatures of interest
- For example:
  - Searching for PNG image files from phone dump and carving them out

1. What can be carved?
   - Disks & Disk Images
   - Memory
   - Files of unknown formats
     - embedded artefacts
2. Objects that may be recovered:
   - Media files
   - Text files and documents
   - Cryptographic Keys

Why?

- No compatible analysis software
- Directory entries are damaged
- File formats are not known

- Identify the desired block of data
  (File Header and Footer) -> save or export for analysis

- Issues involve
  - Some file formats do not have footers
  - For fragmented files the footer may be in another location
  - Encryption / Compression
  - Corruption – if you are searching for a known signature, then a single bit of corruption will cause a specific search to fail.

Where?

- Disk images – partitions, whole disks, USB drives, SD cards etc.
- Memory – memory captures
- Other files – some file formats allow the embedding of files within them, you may not know the host format, but you may still be able to recover internal files.
- Carving need not necessarily be just for files
  - looking for anything in a known format, searching for the appropriate key string can be done.
  - Credit card numbers, e-mails addresses, telephone numbers, NHS IDs etc.

- Note: the shorter and less unique the search terms the larger the number of false positives.

Example

- Step 1: Identify the File Header
  - FF D8 FF→ Common JPEG file header
- Step 2: Identify the File Footer
  - FF D9→ Standard JPEG end-of-file marker
- Step 3: Reconstruct the File
  - Extract all bytes from the JPEG header (FF D8 FF)
  - Continue extraction until the footer (FF D9)
  - Save the extracted bytes as a new file
- Step 4: Interpret the Result
  - Possible outcomes:
    1. Image opens correctly → likely successful carve
    2. Image partially corrupted → file may be fragmented or partially overwritten
    3. Image does not open → header/footer present but internal structure damaged

190624wu-95-copy~5 (1).jpg

00000002	00 01	02 03	04 05	06 07	08 09	0a 0b	0c 0d	0e 0f	
00000000	ff d8	ff e0	00 10	4a 46	49 46	00 01	01 01	00 60	ÿØÿà..JFIF.....`
00000010	00 60	00 00	ff e1	02 be	45 78	69 66	00 00	4d 4d	.`..ÿá.¾Exif..MM
00000020	00 2a	00 00	00 08	00 02	87 69	00 04	00 00	00 01	.*.....‡i.....
00000030	00 00	01 32	ea 1c	00 07	00 00	01 0c	00 00	00 26	...2ê.....&
00000040	00 00	00 00	1c ea	00 00	00 01	00 00	00 00	00 00	.....ê.....
00000050	00 00	00 00	00 00	00 00	00 00	00 00	00 00	00 00	.....
00000060	00 00	00 00	00 00	00 00	00 00	00 00	00 00	00 00	.....
00000070	00 00	00 00	00 00	00 00	00 00	00 00	00 00	00 00	.....
00000080	00 00	00 00	00 00	00 00	00 00	00 00	00 00	00 00	.....

00110300	12 54	54 51	40 05	43 45	15 48	02 8a	28 a6	01 45	.TTQ@.CE.H.Š(¦.E
00110310	14 50	04 b8	15 15	14 52	40 14	51 45	30 0a	28 a2	.P.¸...R@.QE0.(¢
00110320	80 1d	81 52	03 8a	28 a4	c0 0f	5a 6b	75 a2	8a 90	€..R.Š(¤À.Zku¢Š.
00110330	14 74	a5 a2	8a 00	28 a2	8a 00	93 75	37 cc	f6 a2	.t¥¢Š.(¢Š.“u7Ìö¢
00110340	8a 00	63 9d	ad 81	4d a2	8a 00	64 5d	0d 42	b4 51	Š.c...M¢Š.d].B´Q
00110350	4f a8	11 91	8a 18	d1 45	50 30	c9 a2	8a 28	03 ff	O¨.‘Š.ÑEP0É¢Š(.ÿ
00110360	d9 ..	.. ..	.. ..	.. ..	.. ..	.. ..	.. ..	.. ..	Ù

Questions?

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Useful sources

- https://www.file-extensions.org/
- https://www.garykessler.net/library/file_sigs.html
- Brett Shavers, John Bair, 2016. Hiding Behind the Keyboard, Chapter 7 - Antiforensics, Syngress, Pages 153-172, ISBN 9780128033401

Summary

- Metadata is a very useful source of information and can be found as system and application metadata
- The file signature analysis allows us to identify disparities between the file's signature and extension and often whether a user altered the signature of a file to hide data
- Data carving is a technique that allows the investigator to extract and retrieve data that may hidden in various places

References

- Buchholz and Spafford, (2004), *On the role of file system metadata in digital forensics, Digital Investigation, vol 1*
- Casey, E. (2011). Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet. Academic Press Inc (London) Ltd. ISBN 978-0-12-374268-1
- Eoghan Casey, Curtis W. Rose, Handbook of Digital Forensics and Investigation, 2010, Heidelberg.
- Useful Sources:
  - The Source of File Extensions Information
  - GCK'S File Signatures Table
  - Brett Shavers, John Bair, 2016. Hiding Behind the Keyboard, Chapter 7 - Antiforensics, Syngress, Pages 153-172, ISBN 9780128033401